

## What Did I Do

For this assignment, I explored two online security tools, ShieldsUP! from Gibson Research Corporation (GRC) and Have I Been Pwned (HIBP) to look at two different aspects of cybersecurity. ShieldsUP! was used to see how my network appeared externally while HIBP was used to see whether my email addresses and passwords had appeared in known data breaches. To start as someone who is new to cybersecurity, I found it essential to understand what each tool was actually checking before trying to interpret the results. The tools do not simply tell someone whether their computer is safe or unsafe. Instead, they provide information about specific security conditions that can then be used to identify possible risks.

I started with ShieldsUP! as it provided a way to look at my network from an outside perspective. It is a web based service therefore software did not have to be installed on my computer. Instead, GRC's systems sent network probes from the Internet toward my public IP address and recorded how my network responded. This is different from a tool that scans a computer from inside the network because ShieldsUP! is essentially trying to see what an outside computer can see when it tries to communicate with my network.

To eventually understand the results, I first had to learn the basics and research what a network port is. A port is virtual label in a computer's operating system that controls incoming and outgoing network data to the correct corresponding application or service. I found there are over 65,000 possible Transmission Control Protocol (TCP) ports and many common services traditionally use specific port numbers.

I then ran both the Common Ports test and the All Service Ports test on ShieldsUP!. The Common Ports test focuses on commonly used ports while the All Service Ports test provides a broader test of the first 1,056 ports. GRC explains that the additional ports beyond the traditional service ports are included partly because Windows systems have historically exposed services in portions of the higher numbered port range (Gibson Research Corporation [GRC], n.d.). The Common Ports scan is shown in Appendix A, Figure A1 and the All Service Ports scan is shown in Appendix A, Figure A2.

From the results, one of the terms I had to dive into was "stealth." A port can generally be described as open, closed, or stealthed based on how it responds to an outside request. An open port responds in a way that indicates a service may be listening. A closed port responds but indicates that there is not a service accepting the connection. A stealthed port does not provide a response to the probe. GRC explains that stealthing can make a computer or network less visible to outside systems attempting to scan it (GRC, n.d.).

The second tool, Have I Been Pwned, works differently from ShieldsUP! in that it cross references submitted information against its collection of known breach data. An email address appearing in a breach does not automatically mean that the person's email account was hacked. It may mean the email was exposed through another company or website where the person had an account. I checked a total of three email addresses. I also attempted to check a phone number as required by the assignment. However, the current public HIBP website no longer provides the phone number search option.

Finally, I checked five passwords using HIBP's Pwned Passwords feature. This feature checks whether the password itself has appeared in known breach data and reports how many times it has been observed. Importantly, the number does not represent how many times an individual's account was hacked. For example, a result of 66 means that the password itself appeared 66 times in the breach data, not that my account was compromised 66 times. Additionally, HIBP also uses a privacy protecting method when checking passwords rather than simply sending the full submitted password to the service.

## **What Are the Results**

In general, the ShieldsUP! results were mostly positive. Both the Common Ports and All Service Ports scans produced an overall stealth result. This means that the tested ports did not respond to GRC's external send an Echo Request and determine whether a destination responds. My network responded to GRC's request meaning it was externally visible to this particular type of probe. GRC notes that systems can often be configured to ignore these requests to reduce visibility (GRC, n.d.). This should not be interpreted as evidence that the computer was compromised. If reducing external visibility is a priority, blocking unsolicited ping requests could be considered after determining it does not interfere with actual network troubleshooting.

The HIBP results were more concerning. All three email addresses appeared in known breach databases. The first email address appeared in six breaches, the second in three, and the third in two. These results indicate that the email addresses were included in information associated with known breaches. However, they do not prove that the corresponding email accounts were compromised. The specific breach information included email addresses, names, phone numbers, and physical addresses. This creates a significant privacy and social engineering concern because attackers can combine this information to make phishing emails, calls, or text messages appear more legitimate.

The password results presented the greatest cybersecurity concern. All five passwords had previously appeared in known breach data. The first password had been observed 33 times, the second 33 times, the third 66 times, the fourth 27 times, and the fifth 41 times. The available screenshot showing the 66 times result is included in Appendix B, Figure B1.

The 66 times result requires careful interpretation. It does not mean that the individual's account was breached 66 times, nor does it mean the password appeared in 66 breaches involving that person. It means the password itself was found 66 times in the breach data analyzed by HIBP. The same interpretation applies to the other password counts.

Although these numbers do not prove individual account compromise, they are still a significant security concern if the passwords are currently being used. A password that has appeared in breach data should no longer be treated as confidential. The greatest risk would occur if one of

these passwords were reused across multiple accounts as attackers can use previously exposed credentials in automated attacks.

### **What Did I Learn**

The biggest lesson I learned from this assignment is that cybersecurity tools and results need to be understood before they are used and reported. Each tool measures a particular aspect of security, and the results must be interpreted according to what was actually tested to be used effectively in mitigating cybersecurity risks.

The ShieldsUP! exercise helped me understand the relationship between ports, services, firewalls, and attack surface. I learned that services listening on ports can provide potential entry points for attackers while firewalls can limit unwanted inbound connections. The fact that my tested ports were stealthed indicates that the network had good protection against the specific external probes performed by ShieldsUP!. However, the successful ICMP response demonstrated that the network was still visible to a basic ping request. This showed me that a network can have strong port protection while still providing some information to an outside system.

As a result, I would change my password behavior by using unique passwords for different accounts rather than reusing credentials. I would also consider using a password manager to generate and store strong, unique passwords and enable multifactor authentication, particularly for important accounts such as email and financial services. A password should not be considered safe simply because an account has never reported a compromise.

Finally, a big takeaway is that information such as names, addresses, telephone numbers, and email addresses can create security risks. While these types of information may not seem as sensitive as a password, they can be combined to make social engineering attacks more convincing which enhances the chance of an actual cybersecurity breach.